PROJECT DESIGN PHASE-II
DATA FLOW DIAGRAM & USER STORIES

Team ID: SWTID-2026-1398	**Date:** 30 September 2026
Project Title: Import Data Using Transform Maps	**Maximum Marks:** 4 Marks

1. Project Description & Overview

This project models the automated import process of employee information into ServiceNow using Import Sets and Transform Maps. The spreadsheet contains key employee details including Employee ID, Name, Email, Department, and Location. The uploaded data is temporarily stored and processed in an Import Set Staging Table prior to validation and transformation. A Transform Map then maps these staging fields to the corresponding target fields in the Employee Master Table. Using Employee ID as the unique coalesce key, existing records are updated while new employee entries are inserted seamless into the database.

2. Data Flow Diagram (DFD) Specification

A Data Flow Diagram (DFD) represents the flow of information within the system, detailing inputs, outputs, processes, and data stores.

• External Entities:

- **System Administrator / HR Admin:** Uploads source spreadsheet files and monitors import execution/logs.
- **Target Employee Table (sys_user):** The primary target repository for verified employee records.

• Processes:

- **1.0 Load & Parse Spreadsheet:** Receives the raw spreadsheet containing Employee ID, Name, Email, Department, and Location, loading raw rows into staging.
- **2.0 Validate & Stage Data:** Validates field formats and stores structured data temporarily in the Import Set Table.
- **3.0 Transform & Coalesce Match:** Executes field mapping rules and matches Employee ID as a unique coalesce identifier against existing records.
- **4.0 Execute Record Insert/Update:** Updates existing employee profiles if matching Employee ID is found; inserts a new record otherwise.

• Data Stores:

- **D1: Import Set Staging Table:** Temporary data store holding raw incoming spreadsheet rows before transformation.
- **D2: Employee Master Table:** Permanent target database storing verified active employee information.

3. User Stories & Acceptance Criteria

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
System Administrator	Data Import & Staging	USN-1	As a System Admin, I want to upload an Excel spreadsheet containing employee details (ID, Name, Email, Dept, Location) into ServiceNow.	The uploaded data is stored correctly in the temporary Import Set Table without loss of fields.	High	Sprint-1
System Administrator	Field Mapping	USN-2	As a System Admin, I want to configure a Transform Map to map staging fields to target Employee table fields.	All fields (Employee ID, Name, Email, Department, Location) map accurately to corresponding target fields.	High	Sprint-1
System Administrator	Data Deduplication	USN-3	As a System Admin, I want the system to use Employee ID as a coalesce key to match existing records.	Existing records matching the Employee ID are updated with new spreadsheet data rather than duplicated.	High	Sprint-1
System Administrator	Record Insertion	USN-4	As a System Admin, I want unmatched employee records to be inserted as new entries.	New records with unique Employee IDs are created in the target Employee table.	High	Sprint-1
HR Administrator	Exception Handling	USN-5	As an HR Admin, I want to view import log errors for failed or invalid rows.	Skipped or failed records generate actionable error messages in the import log.	Medium	Sprint-2

HR Administrator	Data Verification	USN-6	As an HR Admin, I want to verify that transformed employee data reflects accurately in employee profiles.	Employee profiles show updated Email, Department, and Location upon import completion.	Medium	Sprint-2